

Experiments B and D

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1 Setup

The environment consists of a finite set of social states and a finite set of actions. At each timestep, the community encounters one state, and agents either act directly or react to the actions of others.

The community contains heterogeneous agents. Each agent has a reward structure over state-action pairs, so agents do not value all behaviors identically even when they observe the same event.

A more detailed discussion on the reputation learning algorithm is in the Modeling section.

2 Experiment B: Reputation Scaling

This experiment is a replication of the same experiment run by Alex, using a different version of implementation.

2.1 Environment and configuration

The reputation scaling experiment builds on the personal utility convergence experiment by increasing γ and studying its effects on norm emergence.

Fixed parameters:

- 100 agents
- 10 social states
- 2 actions per states
- The reward function uses a shared-base Gaussian structure with agent heterogeneity:
 - For each state-action pair (s, a) , the model first draws a shared base payoff $m(s, a)$ from a Gaussian distribution.
 - Then, for each agent i , it draws an agent-specific reward $r_i(s, a)$ around that shared base value.
 - Formally,
$$m(s, a) \sim \mathcal{N}(\mu_{\text{base}}, \sigma_{\text{base}}), \quad r_i(s, a) \sim \mathcal{N}(m(s, a), \sigma_{\text{agent}})$$
 - Reward values are then clipped to a positive range.
 - In the implementation used here, the parameters are

$$\mu_{\text{base}} = 0.5, \quad \sigma_{\text{base}} = 0.08, \quad \sigma_{\text{agent}} = 0.1$$

with clipping to $[0.01, 2.5]$.

- This means agents face correlated but non-identical incentives: they tend to agree on which actions are favorable in a given state, but still differ in the exact payoff magnitudes they experience.
- $\kappa = 0$ to disable the status incentives and isolate the effect of the reputation incentive of γ
- **Static** role switching every 3000 timesteps
- 50000 total timesteps

Role updates occur more slowly than action updates, which is important for interpretation. Actions and observed rewards evolve continuously, while role changes occur at scheduled update time points. This separation of time scales means that social influence doesn't change at every timestep; instead, agents accumulate reputation evidence and periodically reevaluate whether following someone else is beneficial. In this framework, γ scales the strength of reputation as a motive for following.

Variable parameter:

We varied $\gamma \in \{0, 1, 1.5, 2, 2.25, 2.5, 2.75, 3, 3.5, 4, 4.5, 5\}$. $\gamma = 0$ is the baseline where we expect no agents to optimize reputation.

2.2 Expected results

1. As γ increases, the opinion leader should have more followers.
2. $\exists \gamma$ such that with all other parameters held fixed and we run the system for enough timesteps, a single opinion leader will emerge in the community with $n - 1$ followers where n is the total number of agents.

2.3 Observed results

2.3.1 Maximum number of followers of a single agent over time

The main figure we tracked is the maximum number of followers of a single agent over time. This number can be understood as the number of agents who participate under the dominant norm.

Table 1: Eventual Follower Count for Top Agent with Different γ values (with $\kappa = 0$), Static Switching

γ	0	1	1.5	2	2.25	2.5	2.75	3	3.5	4	4.5	5
Followers	0	0	0	0	0	0	2	3	9	42	99	99

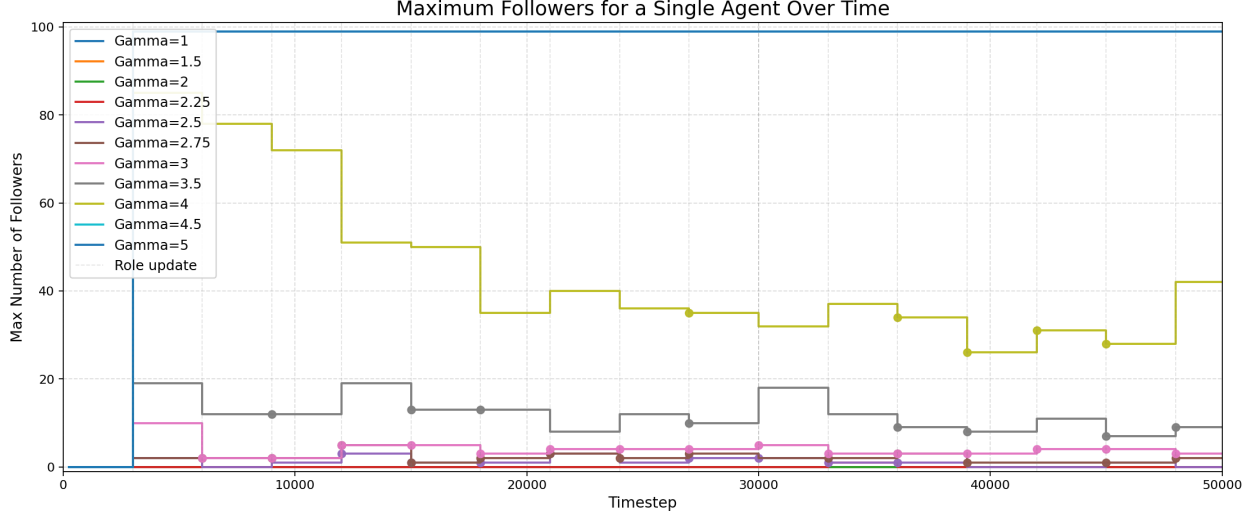


Figure 1: Maximum number of followers of a single agent over time for different γ 's. Static role switching. Blips show the top influencer switching.

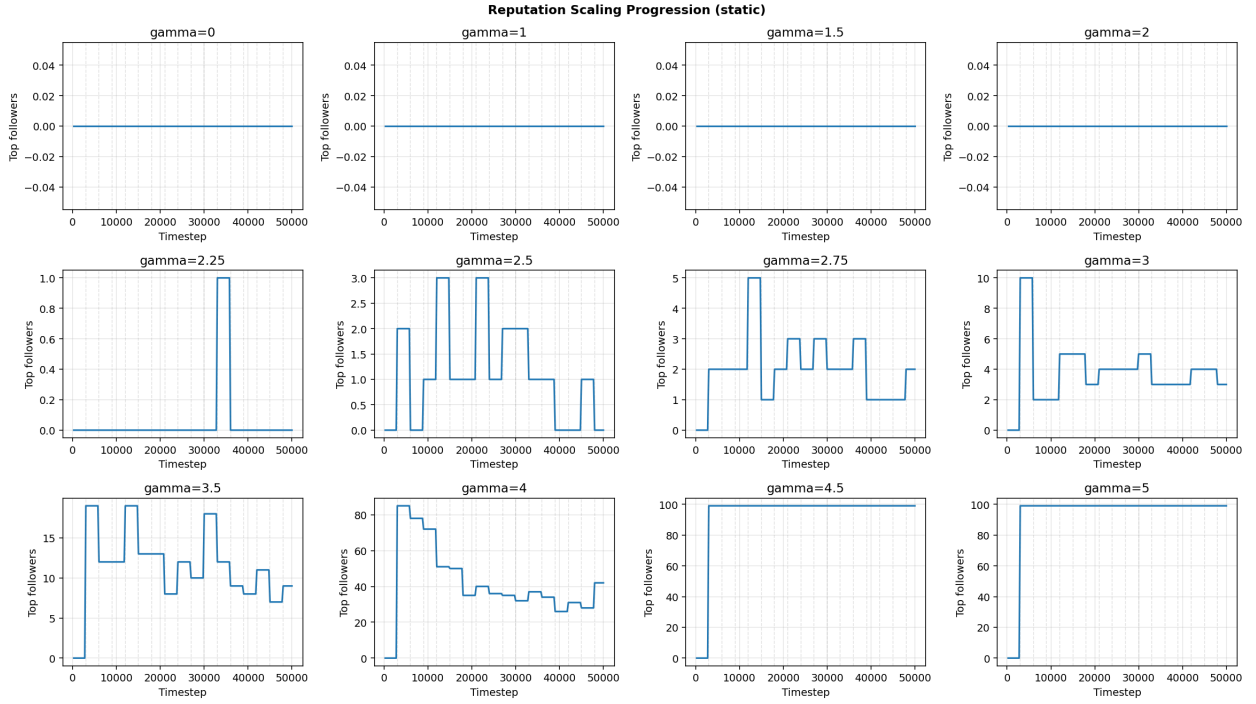


Figure 2: Trajectory of max number of followers faceted by γ

From Table 1, it's clear that the maximum followers increase monotonically as gamma increases and other parameters held fixed, which is consistent with the first hypothesis. Specifically, the number of agents who follow the opinion leader increases from 0 at $\gamma = 0$ to the single digits at $\gamma \in \{2.75, 3, 3.5\}$ to 42 at $\gamma = 4$. The number of participants of the norm reaches 99 at $\gamma = 4.5$. Note that these numbers are not the maximum over timesteps, they are the maximum over all agents who has followers and computed at the last timestep. The effect of γ is nonlinear. From

figure 1 and figure 2, we see that when $\gamma \leq 3$, the reputation scaling effect is minimal, and there are hardly any followers even for the agent with most followers. We start to observe some substantial change to the following behaviour at $\gamma = 4$, where over the top agent garners over 80 followers at the first role switching time point, but the follower base is not stable and doesn't last. For $\gamma = 4.5, 5$, at 3000 timesteps (the first role update), one agent gains 99 followers through direct and chain following. The follower base is stable and the top agent doesn't change.

The exact γ that produces single opinion leader with $n - 1$ followers is different from what Alex obtained, but qualitatively, the results are replicable.

2.3.2 Per-agent personal utility vs. reputation trajectories

Per-agent trajectory of PU and reputation are tracked. The plots for the first 32 agents for $\gamma = 5$ are displayed.

Use agent 0 as an example:

— PU estimate — gamma * reputation — Role update

Experiment B traces: gamma=5 (full)



For agent 0, the blue line is the agent’s learned personal-utility estimate, while the red line is $\gamma * s_i(L_i, t)$, the current reputation value of the agent’s highest-reputation target. In the per-timestep trace data, agent 0 begins with $PU = 0.0221$ and $\gamma * reputation = 0.3175$ at $t=1$, corresponding to a raw selected reputation of 0.0635. By $t=50000$, the same agent has $PU = 0.4833$ and $\gamma * reputation = 0.7353$, which means the raw selected reputation is 0.1471. Personal utility converging near 0.48 is consistent with rewards centered near 0.5, while a modest raw reputation signal becomes dominant once multiplied by $\gamma = 5$.

The red line can continue to move late in the simulation even though agent 0 ends up following agent 38, because the trace records the reputation value of the agent’s currently selected highest-reputation target rather than the reputation of the already-followed leader. Early negative or volatile red values reflect the fact that reputation estimates are learned social signals that can begin near zero.

2.4 Discussion

This experiment examines whether reputation-based social evaluation is sufficient to generate coordinated behavior and norm emergence, without centralized control.

The experimental results provide us the proof that larger values of γ lead to stronger motivation to optimize reputation, stronger imitation of the opinion leader, and a more stable follower base. By contrast, when γ is small, the following dynamics are similar to the no social pressure baseline, where agents do not coordinate.

Notice that in this experiment, κ is kept at 0 – there are no status incentives for anyone, so the opinion leader will always optimize personal utility instead of status. So even though large γ produces a stable norm, this norm is not necessarily welfare optimal.

2.4.1 Seed Robustness

The displayed results are for $seed = 0$. The same experiment was run across 10 seeds (0 to 9 inclusive) to account for stochasticity in activation and learning. 6 out of 10 seeds resulted in a single leader with 99 followers. 3 out of 10 seeds had monotonic increase in the number of followers as γ increases. 1 out of the 10 seeds ends in multi-leader stalemate.

3 Experiment D: Opinion Leader Perturbation

In the Problem Formulation section, we identified one of the causes for norm rupture: opinion leader corruption. When the opinion leader’s actions are no longer beneficial to its followers and hence the community, the followers stop following because they are better off optimizing personal utility.

This experiment provides empirical evidence of what happens to the community after norm rupture, by letting the multi-agent system reach convergence, perturbing the leader so that it chooses actions that dissatisfy its followers, and observing whether we have a period of "anomie" followed by re-emergence of a new leader.

3.1 Environment and configuration

Fixed parameters:

- 100 agents

- 10 social states
- 2 actions per states
- The reward function uses a shared good/bad structure with preserved agent heterogeneity:
 - For each state s , one action $\hat{g}(s)$ is designated as the good action shared across the population.
 - For each agent i , rewards are drawn around a common good/bad base:
$$r_i(s, \hat{g}(s)) \sim \mathcal{N}(1.0, 0.1), \quad r_i(s, a_{\text{bad}}) \sim \mathcal{N}(0.1, 0.1)$$
 - Reward values are clipped to $[0.01, 2.5]$ and adjusted so that the good action remains at least 0.02 above the bad action for every agent in every state.
 - This preserves agent heterogeneity while ensuring that each state has one action that is systematically better than the other for all agents.
- $\gamma = 15$ in the recovery-focused tuned runs reported here
- $\kappa = 2$ to allow status incentives alongside reputation incentives
- **Static** role switching every 3000 timesteps
- 100000 total timesteps. This is an increase from 50000 in experiment B because experiment D involves the whole cycle of norm emergence, perturbation, normless state, and norm re-emergence, thus requiring longer timesteps to see convergence.
- Perturbation is action-only: once the system converges, the current opinion leader is forced onto the bad action in every state for 9000 timesteps, corresponding to 3 role updates
- No direct reputation shock and no forced follower collapse are applied

3.2 Expected results

Assuming the system can reach convergence to a single opinion leader with 99 followers in the first place, we expect:

- The opinion leader starts to lose followers once the perturbation begins. It wouldn't necessarily lose all followers at the first role update epoch following the perturbation, but should be out of followers at some point. This is why we enforce the perturbation for 9000 timesteps – to give followers enough opportunity to learn reputation and react.
- According to the algorithm in Section 7.3 of the math paper (Sharma and Marbach, 2026), at each role update timestep, agents without followers are processed once in sequence and decide whether to optimize reputation by following the agent they currently believe has the highest reputation, provided the reputation condition exceeds both the relevant threshold and their personal-utility estimate. If that condition fails for an existing follower, the agent is removed from the reputation role and later assigned to personal-utility optimization in the same update pass. Therefore, if i decides to unfollow j who is the corrupt opinion leader at e.g., time 3000, agent i is immediately removed from the follower set and will default to personal utility optimization in step 3 of the algorithm. Effectively, the followers who stop following switch back to optimizing personal utility for a while before following another

high-reputation agent. We expect a period of normlessness where no clear opinion leader is present for the community. This is consistent with the concept of "anomie" proposed by Émile Durkheim.

- We expect that with sufficiently long timesteps and the appropriate parameters, the system will reconverge to a dominant norm, under a new leader.

3.3 Observed results

The 100000-step recovery-focused run was carried out on six seeds that converged to a single opinion leader with 99 followers before perturbation. After perturbation, all 6 seeds exhibited complete collapse of the original leader: in every seed, the pre-perturbation leader eventually lost all followers, and every seed entered a normless period in which no dominant opinion leader remained.

However, re-emergence of a new stable leader was only partially observed. Out of the 6 converged seeds, 3 seeds recovered stably under a new leader, while 3 other seeds did not recover by the end of the run. This gives a convergence rate of 1.0, a collapse rate of 1.0, a normless rate of 1.0, and a stable recovery rate of 0.5. The mean normless duration across the six seeds was 43666.8 timesteps, and the mean final top-follower count was 66.3.

Use seed 0 and seed 1 to illustrate the difference between successful and unsuccessful recovery: Figures 4 and 5 show one failed seed and one successful seed. In each figure, the top panel shows the number of followers of the current top leader (blue) together with the number of followers still attached to the pre-perturbation leader (red). The middle panel shows the share of agents currently optimizing personal utility. The bottom panel shows social welfare over time.

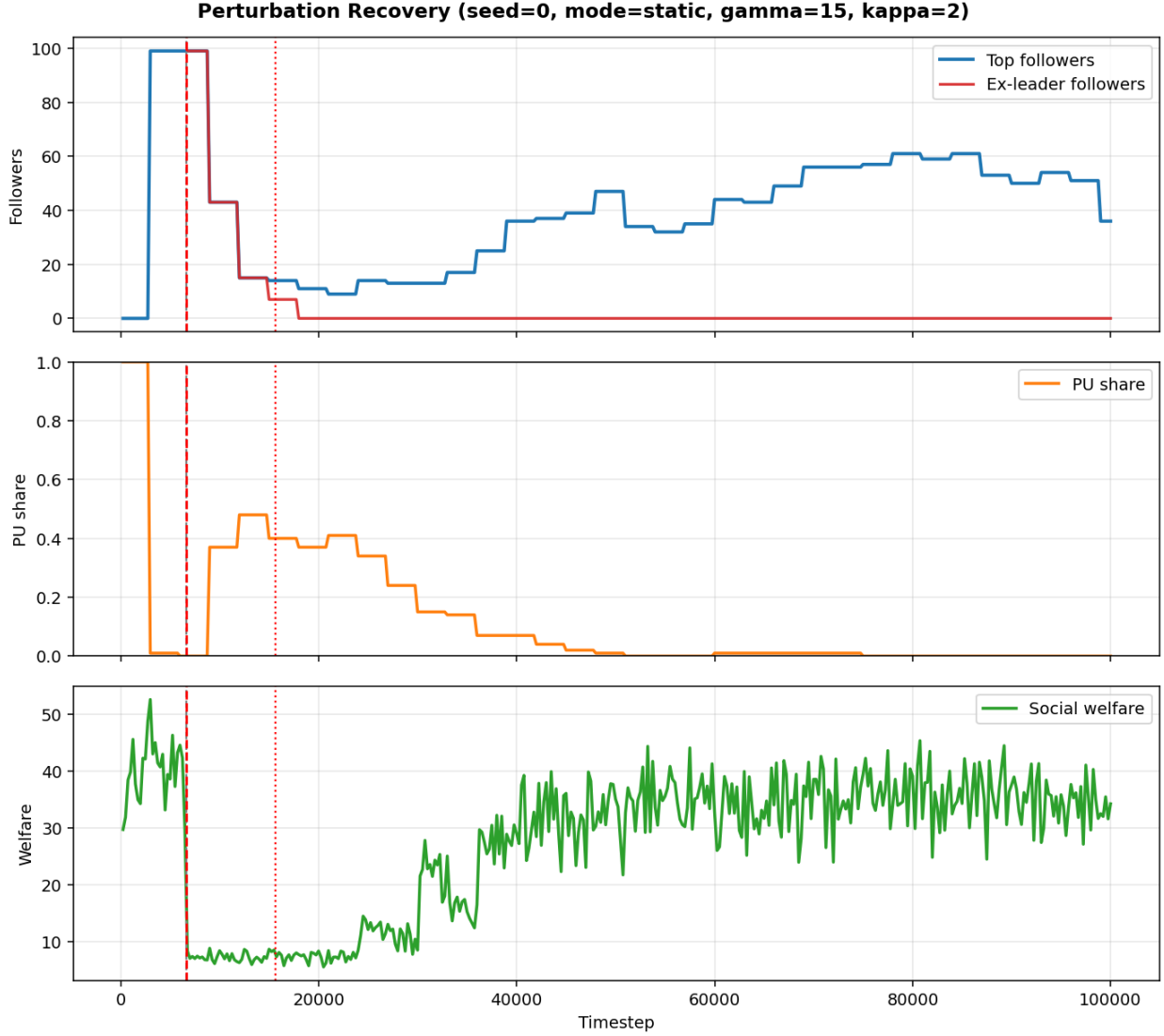


Figure 4: Failed recovery case for Experiment D (seed 0). The top panel shows complete collapse of the original leader but only partial reconcentration around a successor. The middle panel shows that agents temporarily return to personal utility optimization during collapse, but later re-enter reputation-following. The bottom panel shows welfare collapse without stable recovery by the end of the run.

Figure 4 shows a failure of recovery. In the top panel, the original leader’s follower count drops from 99 to 0. A replacement leader does begin to emerge, but it never stabilizes: the largest alternative leader reaches at most 61 followers and the run ends with only 36 followers for the top leader. In the middle panel, the personal-utility share eventually falls back to 0. This means the failure is not due to agents permanently remaining in personal-utility optimization. Rather, agents re-enter reputation optimization but fail to coordinate on a single successor.

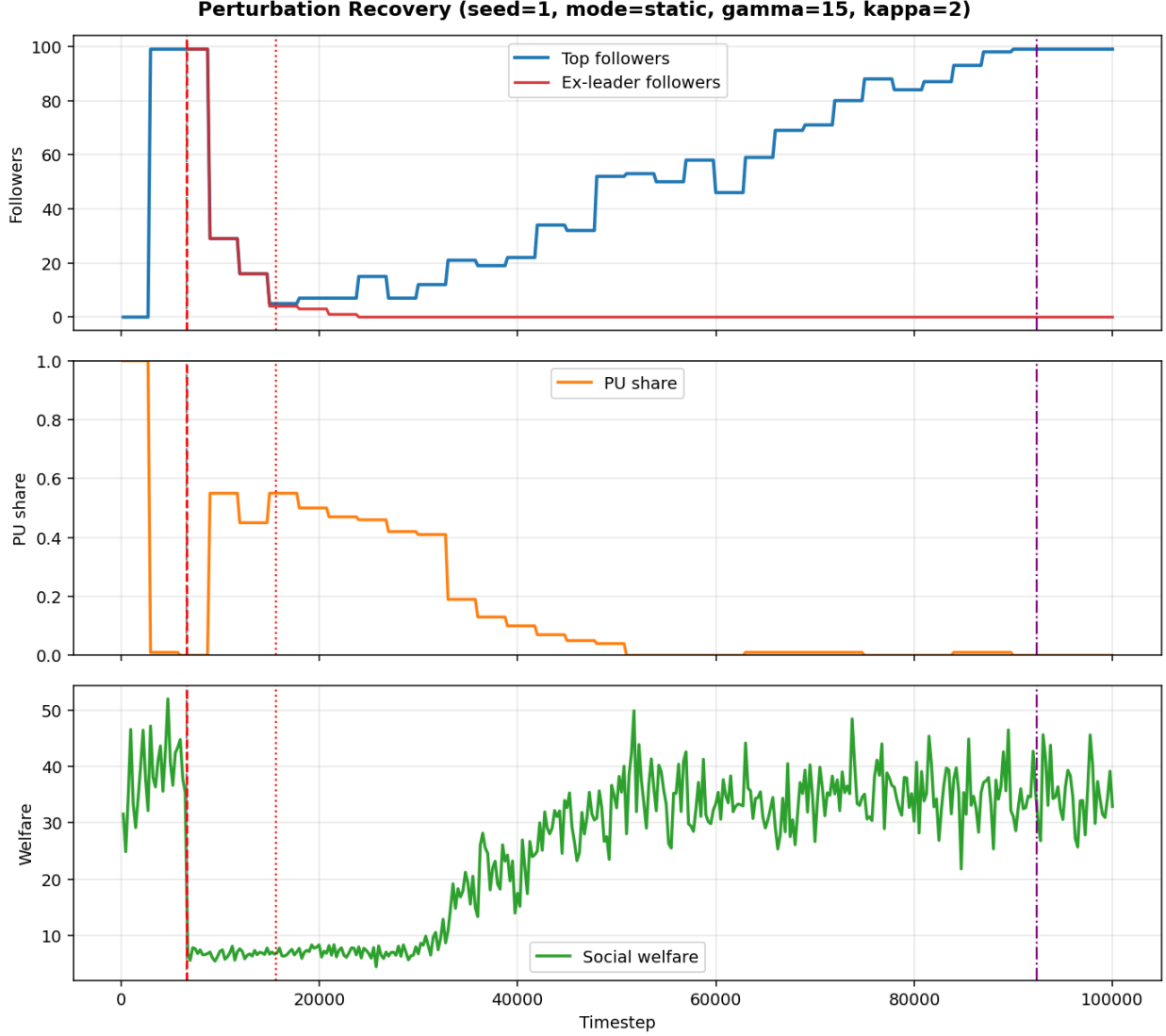


Figure 5: Successful recovery case for Experiment D (seed 1). The top panel shows complete collapse of the original leader followed by gradual concentration around a new leader. The middle panel shows a temporary rise in personal-utility optimization during the normless phase, followed by a return to reputation-following. The bottom panel shows welfare collapse and subsequent recovery under the new leader.

Figure 5 shows the same initial collapse, but successful reconvergence. In the top panel, the pre-perturbation leader again loses all followers, but after a normless period a new leader gradually accumulates followers and eventually reaches 99 followers. In the middle panel, the personal-utility share peaks at 0.55 during the disruption phase and then declines to 0 as the new leader becomes established. In the bottom panel, welfare falls from 35.71 before perturbation to 6.78 during collapse, but then recovers to 35.11 once the new leader has stabilized.

The run-level statistics for seed 1:

- The seed converged at $t_{\text{conv}} = 6599$, the perturbation window was $t = 6600, \dots, 15599$, and the collapse was complete with `drop_fraction = 1.0`.

- The normless period lasted 39000 timesteps.
- Recovery occurred at $t = 92399$, the post-recovery leader was agent 25 (old leader was 74), and the recovery was stable.

3.4 Discussion

The results show that the action-only perturbation is sufficient to destroy an established norm and produce a normless phase. The main remaining source of variation is not whether norm rupture occurs, but whether the system can successfully coordinate on a single (new) leader within the observed timesteps.

The 3 seeds that failed had different failure mechanisms. For seed 4, many agents continue to optimize personal utility rather than reputation. By contrast, seeds 0 and 5 re-enter reputation optimization successfully, but still fail to recover because the follower base does not fully stabilize around a single leader. In other words, some failures arise because agents do not find a new reputation target attractive enough, whereas others arise because they do, but do not coordinate on the same target.

This distinction is important. The experiment suggests that norm re-emergence requires more than dissatisfaction with a perturbed leader. It also requires successful concentration of followers around a single replacement leader. A norm can therefore collapse quickly but remain absent for a long period.